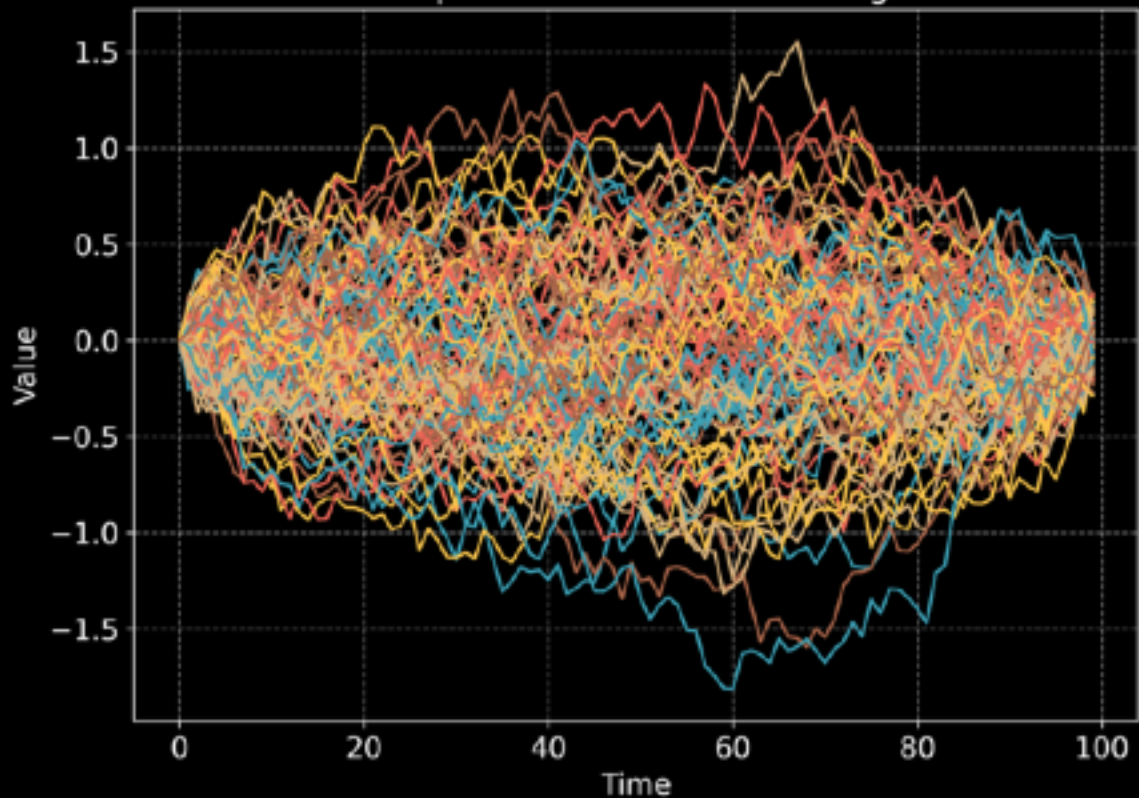


Sample Paths of Brownian Bridge



Schrödinger Problem

- Schrödinger Bridge problem: $\min_{\bar{\pi} \in \bar{\Pi}(\rho_0, \rho_1)} \text{KL}(\bar{\pi} \parallel \bar{\mathcal{K}})$ where $\bar{\mathcal{K}}$ = symmetric Wiener measure.

- Has a unique solution $\bar{\pi}_\varepsilon^\star$

- $\bar{\pi}_\varepsilon^\star$ = mixture of Brownian bridges $\bar{\gamma}_{x,y}^\varepsilon \in \bar{\mathcal{X}}$ ($\sim$ a *fuzzy* geodesic) connecting x, y

- **Connections:**

- As noise level $\varepsilon \rightarrow 0$: Schrödinger Bridge $\rightarrow$ OT on paths, and $\bar{\gamma}_{x,y}^\varepsilon \rightarrow \delta_{\gamma_{x,y}}$
- $\bar{\pi}_\varepsilon^\star$ is related to EOT solution $\pi_\varepsilon^\star$ via Brownian bridges. For discrete measures:

$$\pi_\varepsilon^\star = \sum_{i,j} \pi_{\varepsilon,i,j}^\star \delta_{(x_i, y_j)} \quad \text{and} \quad \bar{\pi}_\varepsilon^\star = \sum_{i,j} \bar{\pi}_{\varepsilon,i,j}^\star \bar{\gamma}_{x_i, y_j}$$

- Intermediate marginal at time t : $\rho_t^\varepsilon = (P_t)_\# \bar{\pi}_\varepsilon^\star \sim$ entropic displacement interpolation



Schrödinger Problem